

Wesley DeMontigny

University of Maryland, College Park

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Graduate Research Assistant, University of Maryland | 2023–Present

- Developed statistical and machine-learning methods for incomplete, heterogeneous scientific data.

Selected Projects:

Cherry Blossom Forecasting Pipeline | [Write-Up](#)

- Won the Most Accurate Prediction award from the 2026 International Cherry Blossom Prediction Competition by fusing mixed-effects models with LSTM-based climate representation learning.
- Pre-trained a stacked LSTM in PyTorch on NASA POWER temperature and NOAA ENSO data, then transferred its latent representations to an MLP to forecast bloom-date residuals under extreme data scarcity (5-day MAE).

Probabilistic Imputation Model for Incomplete Genomic Data | [GitHub](#), [Preprint \(under review\)](#)

- Designed and implemented a probabilistic graphical model in Python (JAX/NumPyro) to impute missing values in high-dimensional binary vectors.
- Achieved $2\times$ higher precision than competing methods at comparable recall, and enabled uncertainty quantification for downstream analysis.

Reversible-Jump MCMC for Regime-Switching Models | [GitHub](#), [Preprint](#)

- Engineered a high-performance C++17 reversible-jump MCMC sampler for a nonparametric regime-switching model with an unknown number of latent states.
- Managed experiments on HPC clusters using Slurm (sbatch, job arrays), validating sampler correctness through simulation-based posterior coverage tests.

The Bayesian Battleship Benchmark | [GitHub](#), [Write-Up](#)

- Designed an efficient Sequential Monte Carlo sampler to approximate the combinatorial posterior over constrained Battleship configurations.
- Benchmarked Bayesian policies (MAP, information gain, Thompson sampling) against frontier and open-source LLMs using Bradley–Terry models and analyzed their decision behavior.

Scientific Software Recovery and Optimization | [GitHub](#), *In Prep*

- Diagnosed and corrected critical defects in a legacy C++ Bayesian MCMC codebase, added automatic proposal tuning that improved sampling efficiency by $>2\times$, and helped restore a stalled research project to a publication-ready state.

Education:

Ph.D. Candidate in Biological Sciences, May 2027, *University of Maryland* GPA: 4.0

- **Focus:** Computational Statistics, Bayesian Inference, and Computational Biology
- **Dissertation Topic:** Imputation Models for High-Dimensional Partially Observed Datasets

B.S. in Biology, 2022, *Georgia College & State University* GPA: 3.95

- **Awards:** Goldwater Scholar (2022) | Commencement Speaker for College of Arts & Sciences

Skills:

- **Languages & Tools:** C++, Python, R, SQL, Linux CLI, Bash, Git, CMake, Docker, Slurm
- **Machine Learning & Probabilistic Frameworks:** PyTorch, PyG, JAX, NumPyro, Scikit-Learn, NumPy, SciPy, pandas, Eigen3, OpenMP, Boost
- **Modeling Expertise:** Bayesian Inference, Graphical Modeling, Latent Variable Modeling, Probabilistic Clustering, Deep Learning, Time Series Analysis